COMP9417

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1 Methodology

The dataset that I choose is already preprocessed and only has continuous feature and the binary one. So, I only apply `StandardScaler` to the continuous data. The dataset has about 70000 rows, so I decide to split it into training, validation and test sets by proportion of $0.4, 0.3$ and 0.3 respectively.

Then, I used the validation set to tuning the hyperparameter for xRFM model based on the paper TALENT hyperparameter distribution. The tuning is done by `GridSearchCV` and due to CPU limitation as well as computation time, only 5 trial of tuning is used. Similar method are also used for the baseline model, xGBoost and MLP. Three models are then being train on the traing set as well as the validation set (for xFRM).

Then, the accuracy and ROC-AUC graph and value is being calculated on test dataset for each model to test its accuracy. The AGOP matrix of the xRFM leaf are also being plotted to present the interpret ability. The PCA loadings and mutual information score are also calculated for feature importance comparison. Permutation importance are attempted but have to put it for later since it take too long.

Then, the training time and test performance of three models with different size are plotted by testing the traing time and accuracy of each model on subsample of various size.

2 Result

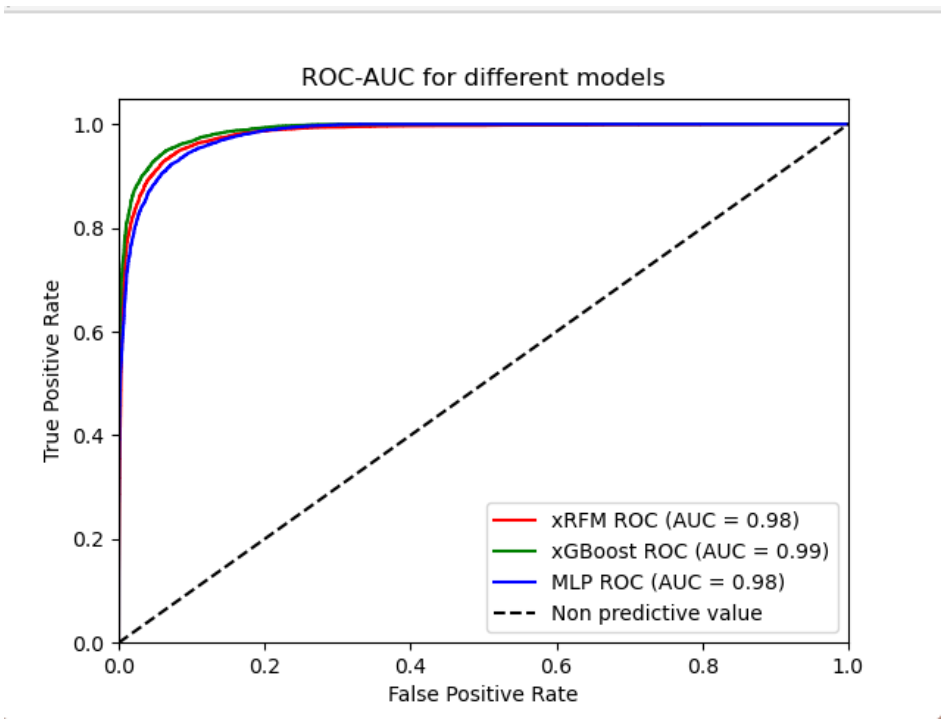
```
Accuracy of xRFM Model: 0.9332380952380952
Accuracy of xGBoost Model: 0.9418095238095238
Accuracy of MLP Model: 0.9248095238095239
```

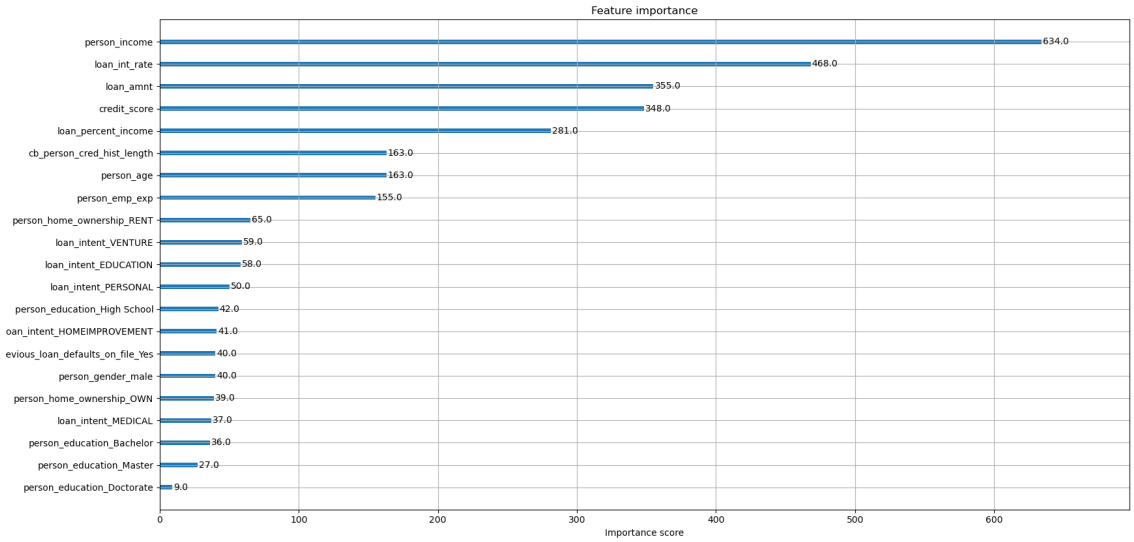
Figure 1: Accuracy on test set

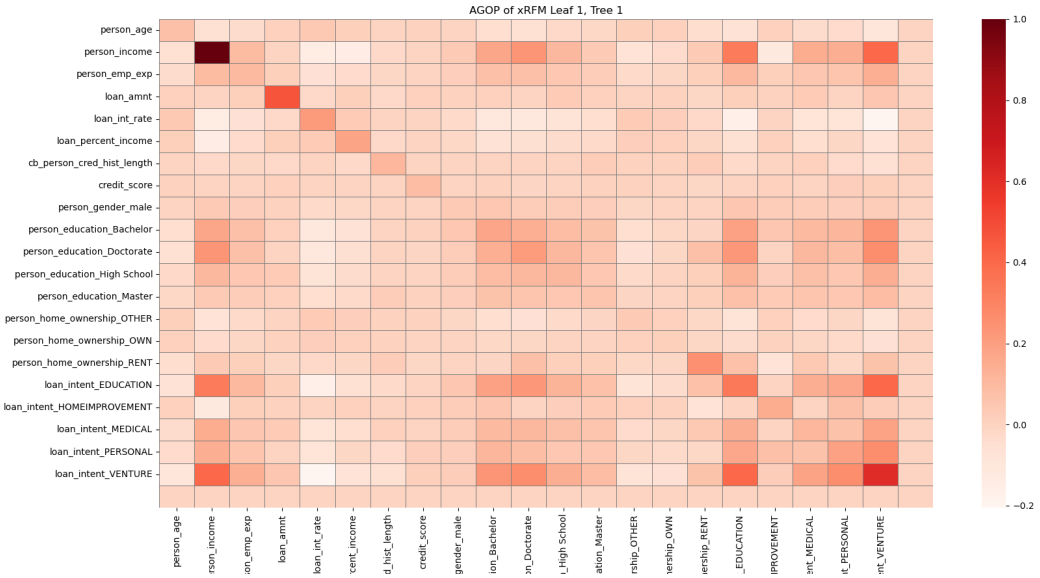
	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	...	PC16	PC17	PC18	PC19	PC20	PC21	PC22	MI score
person_age	0.965951	-0.000681	-0.053985	0.018605	-0.094819	0.005770	-0.015028	0.015165	...	0.009927	-0.020704	0.144899	0.055771	-0.007882	-0.000650	5.214767e-06	0.000000
person_income	0.172091	-0.027776	-0.117840	0.384854	0.645334	-0.114071	0.181512	-0.016042	...	0.002402	0.002106	-0.000341	0.000813	0.000106	-0.000215	-3.919495e-05	0.144925
person_emp_exp	0.951023	-0.023243	-0.049542	0.002997	-0.006967	0.026219	-0.026503	0.023205	...	-0.012708	0.017365	-0.123397	-0.047969	0.007970	0.000235	-7.411796e-07	0.004449
loan_amnt	0.094112	0.867682	-0.189350	0.122032	0.381991	0.056355	-0.005153	0.014103	...	-0.006129	-0.005878	-0.004340	0.004219	0.005316	0.000161	2.944525e-05	0.077955
loan_int_rate	0.018326	0.330232	0.672904	0.013094	-0.226431	0.054250	-0.025776	0.002990	...	-0.001019	0.004775	0.000272	-0.000145	-0.002331	0.000153	-3.004988e-05	0.253283
loan_percent_income	-0.057039	0.893838	-0.116164	-0.314026	-0.248363	-0.023705	0.061319	-0.011277	...	0.007199	0.009032	0.003768	-0.004800	-0.007001	-0.000017	-4.667881e-05	0.172414
cb_person_cred_hist_length	0.923244	-0.003332	-0.058094	0.021226	-0.113292	-0.034155	0.035931	-0.024326	...	-0.001262	0.000942	-0.023636	-0.010268	0.000695	-0.000160	3.013170e-06	0.003572
credit_score	0.267725	0.017167	0.655733	-0.560214	0.432802	0.014166	-0.013950	-0.028472	...	0.005549	-0.007454	0.001827	-0.000762	0.000525	-0.000546	-2.783305e-06	0.003474
person_gender_male	0.016988	-0.034688	-0.035305	-0.005462	0.043760	0.438196	0.168344	-0.046750	...	0.005169	-0.015533	0.000032	-0.003295	0.003921	-0.000132	1.599307e-06	0.008612
person_education_Bachelor	-0.012777	-0.023165	0.008082	-0.023753	0.046547	0.057147	0.011008	0.365922	...	-0.007089	0.098622	0.020007	-0.018227	0.014177	0.002358	-5.580722e-05	0.020894
person_education_Doctorate	0.008703	-0.002242	0.001747	-0.003228	0.003048	0.000118	-0.000777	-0.001746	...	0.000901	-0.009481	-0.002529	0.006533	-0.005358	0.009281	5.099796e-05	0.000000
person_education_High School	-0.000084	-0.033776	-0.071131	0.032481	-0.009269	0.043063	-0.021863	-0.252356	...	-0.005488	0.105969	0.022417	-0.018881	0.015840	0.002272	-4.291174e-05	0.018092
person_education_Master	0.002198	-0.018183	0.009863	-0.022349	0.017585	0.017554	-0.002620	-0.041184	...	-0.039542	0.193632	0.032058	-0.027883	0.019527	0.002728	-3.349430e-05	0.018393
person_home_ownership_OTHER	-0.000017	0.000241	0.000069	0.000666	0.000813	0.000319	-0.000987	0.000121	...	0.000002	0.000558	0.000198	-0.000548	-0.000930	-0.000120	3.770150e-02	0.001417
person_home_ownership_OWN	0.001591	-0.008918	-0.008760	-0.008434	0.005020	0.021903	-0.052917	-0.008713	...	-0.014128	0.026742	-0.052311	0.149024	0.103843	-0.000665	1.700065e-04	0.010102
person_home_ownership_RENT	-0.029429	0.057087	0.066115	-0.076890	-0.174253	-0.055303	0.372677	0.025045	...	0.007583	-0.001829	-0.007397	0.020918	0.014715	0.000141	1.456600e-04	0.023156
loan_intent_EDUCATION	-0.020227	-0.026392	-0.023786	-0.008628	0.023678	0.040536	-0.010207	0.002980	...	-0.070583	0.015074	-0.014347	0.041255	-0.059473	-0.001392	-5.466508e-05	0.020293
loan_intent_HOMEIMPROVEMENT	0.013733	-0.009491	-0.004902	0.008889	0.014454	0.006222	-0.009070	-0.001000	...	0.235182	0.073501	-0.033145	0.054731	-0.073555	-0.001385	-6.579742e-05	0.003806
loan_intent_MEDICAL	0.008218	-0.014059	0.004009	-0.003028	-0.006556	0.009899	0.008798	0.003702	...	-0.054394	0.013131	-0.016553	0.037812	-0.052443	-0.001491	-5.343264e-05	0.002790
loan_intent_PERSONAL	0.011999	-0.020364	-0.017171	-0.005204	0.014522	0.024453	0.003107	0.011257	...	-0.006955	0.019749	-0.018676	0.044167	-0.061758	-0.001363	-2.761428e-05	0.008290
loan_intent_VENTURE	0.004890	-0.018897	-0.019740	-0.008677	0.029578	0.032403	-0.017122	0.003053	...	-0.091452	0.014290	-0.015095	0.036241	-0.068335	-0.001097	-6.839757e-05	0.030194
previous_loan_defaults_on_file_Yes	0.000517	-0.135800	-0.155144	0.023928	0.095079	0.176578	-0.112743	0.043894	...	0.048348	-0.051570	0.009102	-0.016497	0.011764	-0.000522	-9.898093e-06	0.315420

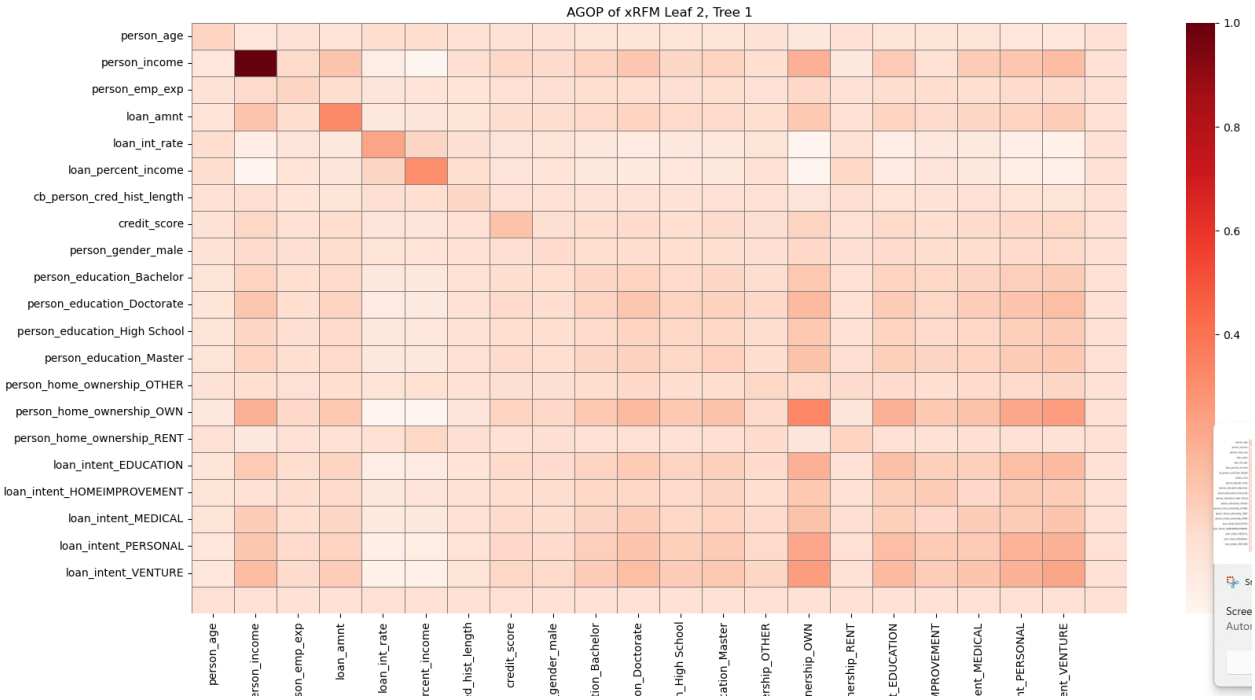
[22 rows x 23 columns]

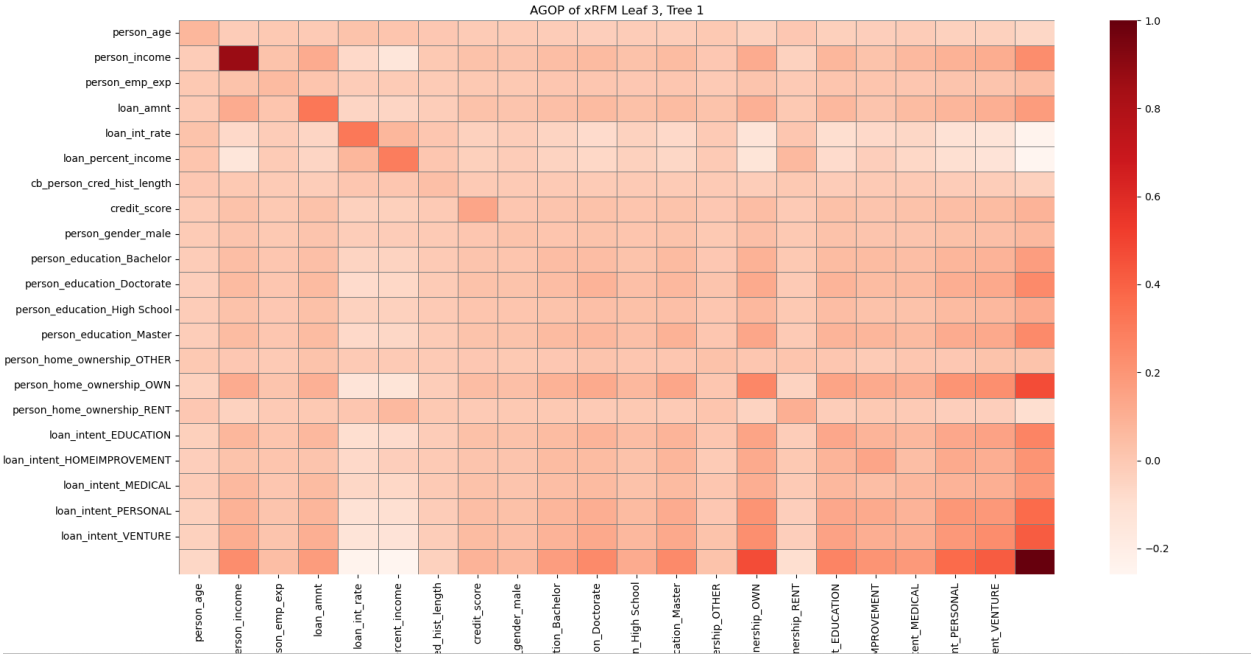
Figure 2: PCA loadings and MI scores


ROC-AUC for different models
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
xRFM ROC (AUC = 0.98)
xGBoost ROC (AUC = 0.99)
MLP ROC (AUC = 0.98)
Non predictive value


Feature importance
person_income 634.0
loan_int_rate 468.0
loan_amnt 355.0
credit_score 348.0
loan_percent_income 281.0
cb_person_cred_hist_length 163.0
person_age 163.0
person_emp_exp 155.0
person_home_ownership_RENT 65.0
loan_intent_VENTURE 59.0
loan_intent_EDUCATION 58.0
loan_intent_PERSONAL 50.0
person_education_High School 42.0
oan_intent_HOMEIMPROVEMENT 41.0
evious_loan_defaults_on_file_Yes 40.0
person_gender_male 40.0
person_home_ownership_OWN 39.0
loan_intent_MEDICAL 37.0
person_education_Bachelor 36.0
person_education_Master 27.0
person_education_Doctorate 9.0
0
100
200
300
400
500
600
Importance score


AGOP of xRFM Leaf 1, Tree 1
person_age
person_income
person_emp_exp
loan_amnt
loan_int_rate
loan_percent_income
cb_person_cred_hist_length
credit_score
person_gender_male
person_education_Bachelor
person_education_Doctorate
person_education_High School
person_education_Master
person_home_ownership_OTHER
person_home_ownership_OWN
person_home_ownership_RENT
loan_intent_EDUCATION
loan_intent_HOMEIMPROVEMENT
loan_intent_MEDICAL
loan_intent_PERSONAL
loan_intent_VENTURE
1.0
0.8
0.6
0.4
0.2
0.0
−0.2


AGOP of xRFM Leaf 2, Tree 1
person_age
person_income
person_emp_exp
loan_amnt
loan_int_rate
loan_percent_income
cb_person_cred_hist_length
credit_score
person_gender_male
person_education_Bachelor
person_education_Doctorate
person_education_High School
person_education_Master
person_home_ownership_OTHER
person_home_ownership_OWN
person_home_ownership_RENT
loan_intent_EDUCATION
loan_intent_HOMEIMPROVEMENT
loan_intent_MEDICAL
loan_intent_PERSONAL
loan_intent_VENTURE
1.0
0.8
0.6
0.4



3 Discussion

Difference principle component will have a different important variable. For PC1, the person age, person employee experience and applicant credit history have the most influence on the principle components. Thus, these variable have a relatively high variance in the dataset.

But the more importance once would be the MI score. From the statistic, the feature with the highest MI score is loan previous default (whether a person be able to pay the loan before due date in the history), the second is loan interest rate and then loan percent income (percent of income able to dedicate to loan), person income and loan amount. This tells us about the approval of loan have high dependency on these factors.

On the leaf 1 RFM, the diagonal of the AGOP suggest that the most important feature for loan approval are person income, loan venture intention and loan amount. The loan venture intention is where xRFM diverges from the MI score. Other important features, but less influence, is loan interest rate, loan percent income, which is where it partially agree with MI score and also, not included in MI score, is the person currently renting a home or own a house and is the loan intent for education.

On the leaf 3 xRFM, it also suggesting the important of loan previous default.